

3(a)	J cartilage ; K smooth muscle ; <i>difference</i> I <i>ref. to</i> other tissues <i>plus one from:</i> cartilage in bronchus but not the bronchioles ; suggestion for smooth muscle tissue (<i>ref. to K</i>) ; e.g. proportionally more smooth muscle in the wall of the bronchiole	3
3(b)(i)	<i>Mycobacterium tuberculosis</i> / <i>Mycobacterium bovis</i> ;	1
3(b)(ii)	antigen binding site(s) ; A variable region	1
3(b)(iii)	any two from: A antigen for TB protein A TB pathogen for pathogen causing TB (immobilised) monoclonal antibody has a binding site which is a complementary shape to the protein secreted by the TB pathogen ; monoclonal antibody on the test line will only bind with protein secreted by the by the TB pathogen ; <i>idea that</i> monoclonal antibodies with tiny gold particle are only held in place if protein secreted by the TB pathogen is present ; AVP ; e.g. <i>suggestion that</i> TB protein is specific to the TB pathogen	2

Question	Answer	Marks
3(b)(iv)	any two from: different shaped, variable region / antigen binding site ; ref. to different primary structure ; ref. to a different tertiary structure ; ref. to different (named) bonds holding the tertiary structure ;	2
3(c)	any three from: provides (long-term) immunity ; <i>context of pathogen entering the body</i> ref to secondary / fast / strong, immune response ; because of presence of increased numbers of specific T-lymphocytes ; memory T-lymphocytes, recognise / bind to / activated by, the foreign antigen ; T- helper cells secrete. cytokine / interleukins ; I cell-signalling molecules example of consequence of increased cytokine release ; e.g. increased phagocytosis / angry macrophages increased, B-lymphocyte / humoral, response enhance T-killer cell response	3
3(d)(i)	endocytosis ; A phagocytosis	1

Question	Answer	Marks
3(d)(ii)	any three from: some T-lymphocytes have receptors with a complementary shape to the antigen on the tumour cell ; antigens on the surface of the tumour cell bind to receptor / <i>ref. to</i> clonal selection ; T-lymphocytes, divide by mitosis / <i>ref. to</i> clonal expansion ; T-killer cells are produced that destroy the tumour cell ; method used by T-killer cell to destroy tumour cell ; e.g. perforin, hydrogen peroxide, granzymes A toxins <i>ref. to</i> B lymphocytes and antibodies ; AVP ; e.g. further detail of method used by T-killer cell <i>ref. to</i> phagocytosis of cancer cells	3